

Task 1:

^ Essentials		JSON View
Resource group (move) ID616OSC-Cameron-RG	Operating system Linux (ubuntu 22.04)	
Status Running	Size Standard B1ls (1 vcpu, 0.5 GiB memory)	
Location Australia East	Primary NIC public IP 52.187.228.188 	
Subscription (move) School of ICT Teaching	1 associated public IPs	
Subscription ID 3be7817b-3c0f-4e30-a68b-7d45c10f0544	Virtual network/subnet labvm-vnet/labvm-Subnet	
	DNS name Not configured	
	Health state -	
	Time created 7/21/2026, 10:51 PM UTC	

Public IP: 52.187.228.188

```
oscstudent@labvm-1: ~  
login as: oscstudent  
oscstudent@52.187.228.188's password:  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1062-azure x86_64)
```

Putty SSH login

```
oscstudent@labvm-1:~$ passwd  
Changing password for oscstudent.  
Current password:  
New password:  
Retype new password:  
passwd: password updated successfully  
oscstudent@labvm-1:~$
```

Changing the password

```

oscstudent@labvm-1:~$ hostname
labvm-1
oscstudent@labvm-1:~$ id
uid=1000(oscstudent) gid=1000(oscstudent) groups=1000(oscstudent),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),118(netdev),119(lxd)
oscstudent@labvm-1:~$ uname -r
6.8.0-1062-azure
oscstudent@labvm-1:~$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G   3.8G   26G   13% /
oscstudent@labvm-1:~$ █

```

Hostname, id, uname -r, df -h /

```

oscstudent@labvm-1:~$ git clone https://github.com/OlayinkaOP/checkpoint1-files.git
Cloning into 'checkpoint1-files'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 4 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
oscstudent@labvm-1:~$ cd checkpoint1-files
oscstudent@labvm-1:~/checkpoint1-files$ ls -la
total 20
drwxrwxr-x 3 oscstudent oscstudent 4096 Aug 20 22:11 .
drwxr-x--- 6 oscstudent oscstudent 4096 Aug 20 22:11 ..
drwxrwxr-x 8 oscstudent oscstudent 4096 Aug 20 22:11 .git
-rw-rw-r-- 1 oscstudent oscstudent 3197 Aug 20 22:11 sys_log.txt
-rw-rw-r-- 1 oscstudent oscstudent  103 Aug 20 22:11 users.csv
oscstudent@labvm-1:~/checkpoint1-files$ █

```

Git clone, ls -la

Task 2:

```
oscstudent@labvm-1:~/checkpoint1-files$ sudo ./audit_users.sh
AUDIT SUMMARY
Records read : 4
Already compliant : 3
Remediated : 12
Failed : 1
```

First Run. Successfully summarized.

```
oscstudent@labvm-1:~/checkpoint1-files$ sudo ./audit_users.sh
AUDIT SUMMARY
Records read : 4
Already compliant : 15
Remediated : 0
Failed : 1
oscstudent@labvm-1:~/checkpoint1-files$
```

Second Run: Already compliant for the first run that was remediated. This still has not fixed the failed one though.

```
oscstudent@labvm-1:~/checkpoint1-files$ sudo ./audit_users.sh
AUDIT SUMMARY
Records read : 3
Already compliant : 15
Remediated : 0
Failed : 0
oscstudent@labvm-1:~/checkpoint1-files$
```

Second Run V2: Fixed the failed one. First problem was the csv header being treated as a normal entry, solved by skipping the header names. The second problem was an invisible string present in the beginning of the csv record. The following lines removed the problems.

cat -A users.csv found the invisible string (Byte Order Mark BOM – shows how a file is encoded – removed with those characters because they are the hex representation of it – this was very new and confusing)

```
# remove invisible char from username causing failure
username="${username#'\u0000'}"
# ignore csv header
[[ "$username" == "username" && "$groupname" == "groupname" && "$shell" == "shell" ]] && continue
```

```
oscstudent@labvm-1:~/checkpoint1-files$ sudo usermod -s /bin/sh rob
oscstudent@labvm-1:~/checkpoint1-files$ getent passwd rob | cut -d: -f7
/bin/sh
oscstudent@labvm-1:~/checkpoint1-files$ sudo chmod 755 /hom/andy
chmod: cannot access '/hom/andy': No such file or directory
oscstudent@labvm-1:~/checkpoint1-files$ sudo chmod 755 /home/andy
oscstudent@labvm-1:~/checkpoint1-files$ stat -c '%U %a' /home/andy
andy 755
oscstudent@labvm-1:~/checkpoint1-files$
```

Intentional Breakages before the Third Run.

```
oscstudent@labvm-1:~/checkpoint1-files$ sudo ./audit_users.sh
AUDIT SUMMARY
Records read : 3
Already compliant : 13
Remediated : 2
Failed : 0
oscstudent@labvm-1:~/checkpoint1-files$
```

Third Run: showing the breakages we made were saved, and then remediated by the script.

```

[2026-08-21 00:34:37] FIXED: created group 'admin/lecturer'
[2026-08-21 00:34:38] FIXED: added 'rob' to 'admin/lecturer'
[2026-08-21 00:34:38] COMPLIANT: home directory for 'rob' has correct owner
[2026-08-21 00:34:38] FIXED: home directory for 'rob' had wrong permissions '7
[2026-08-21 00:34:38] FIXED: created 'ola' with shell '/bin/bash'
[2026-08-21 00:34:38] FIXED: created group 'lecturer'
[2026-08-21 00:34:38] FIXED: added 'ola' to 'lecturer'
[2026-08-21 00:34:38] COMPLIANT: home directory for 'ola' has correct owner
[2026-08-21 00:34:38] FIXED: home directory for 'ola' had wrong permissions '7
[2026-08-21 00:34:38] FIXED: created 'andy' with shell '/bin/bash'
[2026-08-21 00:34:38] FIXED: created group 'student'
[2026-08-21 00:34:38] FIXED: added 'andy' to 'student'
[2026-08-21 00:34:38] COMPLIANT: home directory for 'andy' has correct owner
[2026-08-21 00:34:38] FIXED: home directory for 'andy' had wrong permissions '
[2026-08-21 00:37:58] ERROR: invalid username 'username'
[2026-08-21 00:37:58] COMPLIANT: 'rob' shell is '/bin/bash'
[2026-08-21 00:37:58] COMPLIANT: group 'admin/lecturer' exists
[2026-08-21 00:37:58] COMPLIANT: 'rob' is a member of 'admin/lecturer'
[2026-08-21 00:37:58] COMPLIANT: home directory for 'rob' has correct owner
[2026-08-21 00:37:58] COMPLIANT: home directory for 'rob' has mode 700
[2026-08-21 00:37:58] COMPLIANT: 'ola' shell is '/bin/bash'
[2026-08-21 00:37:58] COMPLIANT: group 'lecturer' exists
[2026-08-21 00:37:58] COMPLIANT: 'ola' is a member of 'lecturer'
[2026-08-21 00:37:58] COMPLIANT: home directory for 'ola' has correct owner
[2026-08-21 00:37:58] COMPLIANT: home directory for 'ola' has mode 700
[2026-08-21 00:37:58] COMPLIANT: 'andy' shell is '/bin/bash'
[2026-08-21 00:37:58] COMPLIANT: group 'student' exists
[2026-08-21 00:37:58] COMPLIANT: 'andy' is a member of 'student'
[2026-08-21 00:37:58] COMPLIANT: home directory for 'andy' has correct owner
[2026-08-21 00:37:58] COMPLIANT: home directory for 'andy' has mode 700
[2026-08-26 02:44:17] ERROR: invalid username 'username'
[2026-08-26 02:44:17] COMPLIANT: 'rob' shell is '/bin/bash'
[2026-08-26 02:44:17] COMPLIANT: group 'admin/lecturer' exists
[2026-08-26 02:44:17] COMPLIANT: 'rob' is a member of 'admin/lecturer'
[2026-08-26 02:44:17] COMPLIANT: home directory for 'rob' has correct owner
[2026-08-26 02:44:17] COMPLIANT: home directory for 'rob' has mode 700
[2026-08-26 02:44:17] COMPLIANT: 'ola' shell is '/bin/bash'
[2026-08-26 02:44:17] COMPLIANT: group 'lecturer' exists
[2026-08-26 02:44:17] COMPLIANT: 'ola' is a member of 'lecturer'
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[2026-08-26 02:44:17] COMPLIANT: group 'student' exists
[2026-08-26 02:44:17] COMPLIANT: 'andy' is a member of 'student'
[2026-08-26 02:44:17] COMPLIANT: home directory for 'andy' has correct owner
[2026-08-26 02:44:17] COMPLIANT: home directory for 'andy' has mode 700
[2026-08-26 02:49:48] COMPLIANT: 'rob' shell is '/bin/bash'
[2026-08-26 02:49:48] COMPLIANT: group 'admin/lecturer' exists
[2026-08-26 02:49:48] COMPLIANT: 'rob' is a member of 'admin/lecturer'
[2026-08-26 02:49:48] COMPLIANT: home directory for 'rob' has correct owner
[2026-08-26 02:49:48] COMPLIANT: home directory for 'rob' has mode 700
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[2026-08-26 02:49:48] COMPLIANT: group 'lecturer' exists
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[2026-08-26 02:49:48] COMPLIANT: home directory for 'ola' has mode 700
[2026-08-26 02:49:48] COMPLIANT: 'andy' shell is '/bin/bash'
[2026-08-26 02:49:48] COMPLIANT: group 'student' exists
[2026-08-26 02:49:48] COMPLIANT: 'andy' is a member of 'student'
[2026-08-26 02:49:48] COMPLIANT: home directory for 'andy' has correct owner
[2026-08-26 02:49:48] COMPLIANT: home directory for 'andy' has mode 700

```

CAT VAR LOG: showing timestamps created using the date function inside the log call.

Task 3:

```
oscstudent@labvm-1:~/checkpoint1-files$ ./log_report.sh

Total lines analysed: 32

ACTIVITY BY SERVICE
  14 systemd
  11 sshd
   6 kernel
   2 CRON

PROBLEM LINES: 16 of 32 (50%)

TOP 5 SERVICES BY PROBLEM LINES
   6 systemd
   6 sshd
   4 kernel

oscstudent@labvm-1:~/checkpoint1-files$ cat log_report.txt
=====
SYSTEM LOG REPORT
Generated : 2026-08-26 06:58:54
Log file  : sys_log.txt
=====

Total lines analysed: 32

ACTIVITY BY SERVICE
  14 systemd
  11 sshd
   6 kernel
   2 CRON

PROBLEM LINES: 16 of 32 (50%)

TOP 5 SERVICES BY PROBLEM LINES
   6 systemd
   6 sshd
   4 kernel

oscstudent@labvm-1:~/checkpoint1-files$
```

Script run, both analysis sections + the report

```
oscstudent@labvm-1:~/checkpoint1-files$ ./log_report.sh does_not_exist.txt
Error: file does not exist: does_not_exist.txt
oscstudent@labvm-1:~/checkpoint1-files$ echo $?
1
oscstudent@labvm-1:~/checkpoint1-files$
```

Error handling and non-zero exit code

Service to Investigate First:

Systemd and sshd both have 6 problem lines, however systemd has 14 total lines and sshd has 11 problem lines. Based on the proportions I would look into sshd first.